

Tugas 3

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No. _____
Date _____

Tugas 3

1. $f(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 - \frac{1}{2}x^2 + 2x - 1$

a) Stationer

$$f'(x) = 0$$

$$x^3 - 2x^2 - x + 2 = 0$$

$$x^2(x-2) - (x-2) = 0$$

$$(x-2)(x^2-1) = 0$$

$$(x-2)(x-1)(x+1) = 0$$

$\downarrow$
 $x-2 = 0$
 $x = 2$

$\downarrow$
 $x-1 = 0$
 $x = 1$

$\downarrow$
 $x+1 = 0$
 $x = -1$

b) Naik - Turun

$$f'(x) = 0$$

$$x^3 - 2x^2 - x + 2 = 0$$

$$x = 2, x = 1, x = -1$$

-	+	-	+
-1	0	1	2

$\Rightarrow$

uji dengan 0

 $f'(0) = (0)^3 - 2(0)^2 - 0 + 2$
 $= 0 - 0 - 0 + 2$
 $= 2 \text{ (positif)}$

Dapat disimpulkan $f(x)$ naik pada saat $-1 < x < 1$
 atau $x > 2$ dan $f(x)$ turun pada saat $x < -1$
 atau $1 < x < 2$

c) Cekungan.

$$f'(x) = x^3 - 2x^2 - x + 2 = 0 \quad (\text{titik belok})$$

$$f''(x) = 3x^2 - 4x - 1 = 0$$

$$x = \frac{4 \pm \sqrt{16 - 4(3)(-1)}}{2 \cdot 3} \quad \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

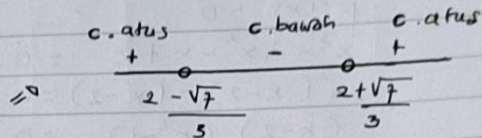
BOSS 6 mm

$$x = 4 \pm \sqrt{\frac{28}{6}}$$

$$= 4 \pm \sqrt{4 \cdot \frac{7}{3}}$$

$$= 4 \pm \frac{2\sqrt{7}}{3} \quad (\text{bagi } 2)$$

$$x = 2 \pm \frac{\sqrt{7}}{3}$$



Jadi $f(x)$ cekung keatas pada saat $x < 2 - \frac{\sqrt{7}}{3}$ atau

$x > 2 + \frac{\sqrt{7}}{3}$, dan cekung kebawah saat

$$\frac{2 - \sqrt{7}}{3} < x < \frac{2 + \sqrt{7}}{3}$$

d) Maksimum, Minimum

e) Maksimum

$$x = 1$$

$$f(1) = \frac{1}{4}(1)^4 - \frac{2}{3}(1)^3 - \frac{1}{2}(1)^2 + 2(1) - 1$$

$$= \frac{1}{4} - \frac{2}{3} - \frac{1}{2} + 1$$

$$= \frac{3 - 8 - 6 + 12}{12}$$

$$= \frac{1}{12}$$

Jadi nilai maks di $\frac{1}{12}$

f) Minimum

$$f''(x) > 0$$

$$x = -1$$

$$f(-1) = \frac{1}{4}(-1)^4 - \frac{2}{3}(-1)^3 - \frac{1}{2}(-1)^2 + 2(-1) - 1$$

$$= \frac{1}{4} + \frac{2}{3} - \frac{1}{2} - 1$$

$$= \frac{3 + 8 - 6 - 12}{12}$$

$$= -\frac{3}{4}$$

Minimum Global

$$x = 2$$

$$f(2) = \frac{1}{4}(2)^4 - \frac{2}{3}(2)^3 - \frac{1}{2}(2)^2 + 2(2) - 1$$

$$= 4 - \frac{16}{3} - 2 + 3$$

$$= 5 - \frac{16}{3}$$

$$= \frac{15}{3} - \frac{16}{3}$$

$$= -\frac{1}{3} \quad \text{Minimum Lokal}$$

e) Ya ada pada $(-1, -\frac{31}{12})$, $(1, \frac{1}{12})$, $(2, \frac{1}{3})$

2.

Misalkan $x + y = 90$

$$y = 90 - x$$

Carilah nilai maksimal dari $x \cdot y^2$

Maka berlaku fungsi :

$$f(x) = x \cdot (90 - x)^2$$

$$f(x) = x \cdot (8100 - 180x + x^2)$$

$$f(x) = 8100x - 180x^2 + x^3$$

$$f'(x) = 8100 - 360x + 3x^2$$

$$f'(x) = 0$$

$$3x^2 - 360x + 8100 = 0 \quad (\text{bagi 3})$$

$$x^2 - 120x + 2700 = 0$$

Cari nilai x dengan rumus abc

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{120 \pm \sqrt{(-120)^2 - 4(1)(2700)}}{2(1)}$$

$$x = \frac{120 \pm 60}{2}$$

$$x = \frac{120 - 60}{2}$$

$$x = 30$$

$$y = 90 - 30$$

$$y = 60$$

Maka nilai $x = 30$ dan $y = 60$

$$3. f(x) = \frac{x^2 - x - 2}{x - 1} \quad (a)$$

(a) asimtot datar tidak ada

(b) asimtot tegak

Jika penyebut $= 0$, maka pembilang $\neq 0$

$$\text{Penyebut } x - 1 = 0$$

$$x = 1$$

uji $x = 1$ ke pada pembilang.

$$= x^2 - x - 2$$

$$= (x - 2)(x + 1)$$

$$(b) x = 1$$

$$= (1 - 2)(1 + 1)$$

$$= -2$$

$-2 \neq 0$, maka asimtot tegak yaitu $x = 1$

$$(b) f(x) = \frac{1}{x^2} - 1$$

(a) asimtot datar

(b) asimtot tegak

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} - 1$$

$$x^2 = 0$$

$$x = 0$$

↓

asimtot tegak $x = 0$

$$= 0 - 1$$

$$= -1$$

asimtot datar $f(x) = -1$

4. $f(x) = x^2 + 1$

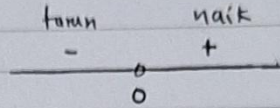
a. Stationer

$$f'(x) = 0$$

$$2x = 0$$

$$x = 0$$

$\Rightarrow$



$\Leftarrow$

Jadi $f(x)$ naik pada saat $x > 0$ dan turun pada $x < 0$.

$$x = 0$$

$$f(0) = 0^2 + 1$$

$$= 1, \text{ titik balik minimal di } (0, 1)$$

b. Kecekungan

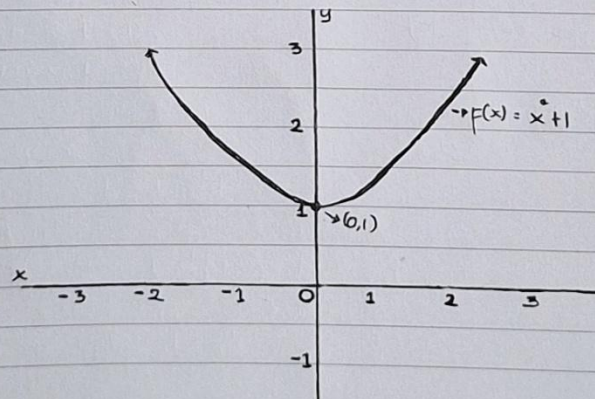
$$f'(x) = 2x$$

$$f''(x) = 2$$

$2 > 0$, maka cekung keatas.

$D(f)$	$f(x)$	$f'(x)$	$f''(x)$	Kesimpulan
$x > 0$		(+)	(+)	$f(x)$ naik, Cekung keatas
$x = 0$	1			Titik balik min $(0, 1)$
$x < 0$		(-)	(+)	$f(x)$ turun, Cekung keatas

c. Gambar grafik Fungsi :



5.

$$a. \lim_{x \rightarrow -1} \frac{x^2 + 5x + 4}{x^2 - 4x - 5}$$

$$\lim_{x \rightarrow -1} \frac{2x + 5}{2x - 4}$$

$$c) \frac{2(-1) + 5}{2(-1) - 4}$$

$$e) \frac{3}{-6} = -\frac{1}{2}$$

$$b. \lim_{x \rightarrow 1} \frac{\ln x}{x^2 - 1}$$

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x}}{2x}$$

$$c) \frac{1}{2(1)}$$

$$e) \frac{1}{2}$$

$$c. \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1}$$

$$e) \frac{1}{\infty}$$

$$e) \frac{0}{1} = 0$$

$$d. \lim_{x \rightarrow \infty} \frac{2x^2 + 5x + 4}{x^2 + 4x - 5}$$

$$\lim_{x \rightarrow \infty} \frac{4x + 5}{2x + 4}$$

$$e) \lim_{x \rightarrow \infty} \frac{4}{2} = 2$$

$$e) \frac{4}{2} = 2$$

